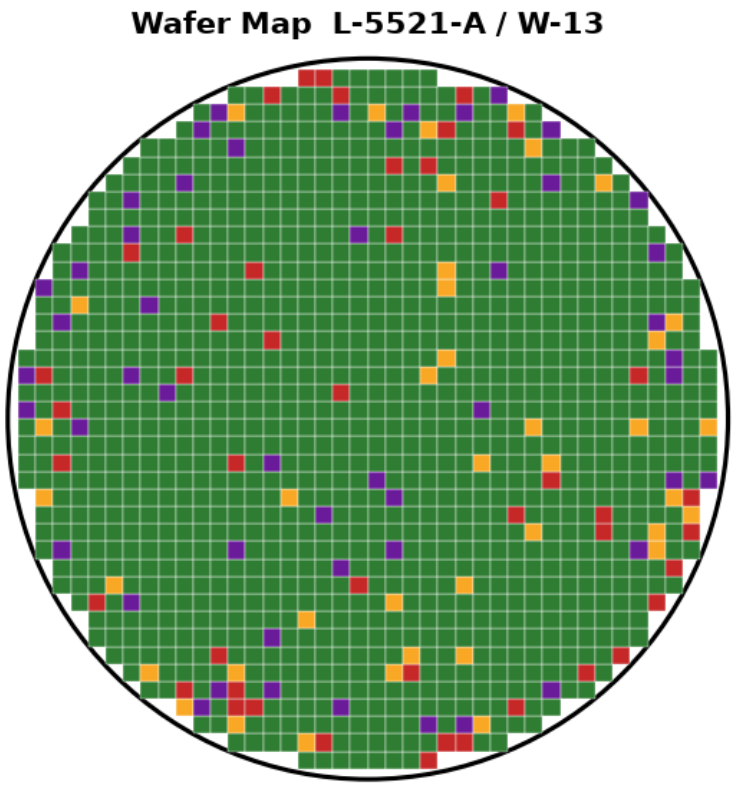
# WAFER PROBE TEST REPORT

Final electrical sort - Product family LX-40 Low-Dropout Regulator

Lot ID ..... L-5534-B
Wafer ID ..... W-02
Product ..... LX-40-50
Test Program ..... LX40_FS_v7.tp
Probe Card ..... PC-0904
Tester ..... T-2000 #2

Operator ..... op-204
Test Temp. ..... 25 °C
Start Time ..... 04/02/2025 09:47
End Time ..... 04/02/2025 11:36
Diameter ..... 200 mm
Notch ..... 270°




Yield: 91.3 %  Good die: 1,204 / 1,319  Edge exclusion: 3 mm

## Bin Summary

| Bin | Description | Count | % |
|---|---|---|---|
| 1 | Pass - all parameters | 1210 | 92.30 |
| 7 | Fail - VOUT out of range | 39 | 2.97 |
| 9 | Fail - Dropout voltage | 24 | 1.83 |
| 12 | Fail - Quiescent current | 28 | 2.14 |
| 15 | Fail - Open/short | 10 | 0.76 |
| | **Total tested** | **1311** | **100.00** |

## Disposition

Wafer released to assembly. Bins 7 and 9 clustered at wafer edge (see map), consistent with known implant non-uniformity. No action required. Inked die: 0 (electronic map only).

## Parametric Results by Test Site (sample of 30 die)

| Die X | Die Y | Site | VOUT [V] | VDROP [mV] | IQ [µA] | Line Reg. [%/V] | Load Reg. [%] | PSRR [dB] | Bin |
|---|---|---|---|---|---|---|---|---|---|
| 10 | 13 | 1 | 5.007 | 196 | 44.8 | 0.029 | 0.33 | 66.0 | 1 |
| 15 | 10 | 2 | 5.005 | 199 | 43.2 | 0.026 | 0.45 | 65.3 | 1 |
| 1 | 11 | 3 | 4.992 | 232 | 46.2 | 0.024 | 0.35 | 68.3 | 1 |
| 10 | -16 | 4 | 5.004 | 214 | 42.0 | 0.009 | 0.24 | 66.9 | 1 |
| -14 | 7 | 1 | 5.018 | 213 | 46.6 | 0.024 | 0.29 | 68.4 | 1 |
| -9 | -3 | 2 | 4.992 | 201 | 38.7 | 0.013 | 0.34 | 65.4 | 1 |
| 10 | -14 | 3 | 4.992 | 218 | 44.0 | 0.013 | 0.38 | 70.3 | 1 |
| 5 | -15 | 4 | 4.999 | 233 | 42.3 | 0.032 | 0.35 | 70.2 | 1 |
| 14 | 8 | 1 | 4.985 | 214 | 37.9 | 0.032 | 0.36 | 69.6 | 1 |
| 11 | -3 | 2 | 5.020 | 215 | 43.8 | 0.015 | 0.39 | 66.2 | 1 |
| 4 | 13 | 3 | 4.989 | 236 | 40.2 | 0.022 | 0.38 | 66.8 | 1 |
| -6 | 5 | 4 | 5.000 | 232 | 47.0 | 0.031 | 0.38 | 67.8 | 1 |
| -9 | -13 | 1 | 5.001 | 230 | 45.9 | 0.027 | 0.32 | 70.1 | 1 |
| -10 | 6 | 2 | 5.008 | 216 | 39.7 | 0.033 | 0.46 | 70.6 | 1 |
| 4 | -16 | 3 | 4.990 | 225 | 43.0 | 0.017 | 0.47 | 65.6 | 1 |
| -7 | -4 | 4 | 4.996 | 218 | 40.4 | 0.029 | 0.43 | 71.0 | 1 |
| -15 | -11 | 1 | 5.002 | 186 | 46.8 | 0.014 | 0.47 | 64.8 | 1 |
| 17 | 10 | 2 | 4.992 | 222 | 43.8 | 0.016 | 0.42 | 65.8 | 1 |
| 17 | -7 | 3 | 5.003 | 219 | 37.6 | 0.016 | 0.31 | 70.5 | 1 |
| -17 | 12 | 4 | 4.997 | 189 | 41.0 | 0.030 | 0.25 | 64.4 | 1 |
| -3 | -4 | 1 | 4.996 | 187 | 37.0 | 0.010 | 0.24 | 64.3 | 1 |
| -9 | 7 | 2 | 5.005 | 187 | 40.4 | 0.012 | 0.29 | 69.8 | 1 |
| -10 | -12 | 3 | 4.999 | 223 | 36.5 | 0.015 | 0.28 | 71.7 | 1 |
| -9 | 10 | 4 | 5.000 | 186 | 40.1 | 0.027 | 0.44 | 69.9 | 1 |
| 4 | -18 | 1 | 5.006 | 187 | 39.7 | 0.025 | 0.43 | 69.8 | 1 |
| 9 | 17 | 2 | 5.005 | 202 | 45.0 | 0.027 | 0.46 | 65.2 | 1 |
| 9 | -11 | 3 | 5.004 | 229 | 39.8 | 0.011 | 0.24 | 67.5 | 1 |
| -3 | -10 | 4 | 4.999 | 194 | 42.2 | 0.032 | 0.36 | 67.8 | 1 |
| -12 | -17 | 1 | 4.991 | 207 | 39.1 | 0.027 | 0.37 | 69.2 | 1 |
| 17 | 14 | 2 | 5.016 | 230 | 42.6 | 0.012 | 0.45 | 64.4 | 1 |
| | | *Limits* | *4.950-5.050* | *<= 250* | *<= 50* | *<= 0.05* | *<= 0.6* | *>= 60* | |